Introduction to Derivatives Valuation

Pablo Marchesi, Xoaquín Amiama

Quantitative Analysts

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$$\sigma^2_{local}(K,t) = \frac{\partial_t C_{mer}(K,t) + rK\partial_K C_{mer}(K,t)}{\frac{1}{2}K^2\partial_{KK} C_{mer}(K,t)}$$

$$dr(t) = \kappa(\theta(t) - r(t))dt + \sigma dW(t)$$

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Introduction

There are two main universes in terms of volume:

1. **Equity: EQ**
2. **Interest Rates: IR**

It is important to note that the EQ underlying is one-dimensional (Spot), whereas the IR underlying is multi-dimensional (Curves).

This leads to more complex models in IR, both mathematically and conceptually.

Introduction

Use of derivatives in banking:

- Hedging: reduce market risks in portfolios and balance sheets.
- Structuring: design solutions for clients (protection, leverage).
- Capital and liquidity management: optimize exposure and limit consumption.

Banks do not usually keep directional positions; that is, when they enter into a derivative with one counterparty, they take the opposite position with another one or unwind and hedge the risk.

EQ: Derivatives

List of some equity derivatives

- Forwards and futures on equities and indices.
- Vanilla options: calls, puts.
- Exotic options: barrier, digital, lookback, quanto.
- Path-dependent products: Asian, barrier.
- Basket products: basket options, worst-of, best-of.
- Structured products: autocallables, barrier baskets, notes.

EQ: Derivatives

Types of options (by exercise style and payoff)

- European: exercised only at maturity.
- American: can be exercised at any time up to maturity.
- Bermudan: exercise on pre-specified discrete dates.
- Asian: depend on the average value of the underlying.
- Barriers: activation or deactivation triggered by a barrier level.
- Digital: pay a fixed amount if a condition is met.

EQ: Valuation Models

Common models

- **Black-Scholes** (BS): constant volatility, closed-form formula for European vanillas.
- **Trees** (binomial/trinomial): useful for American options and early exercise.
- **Monte Carlo** (MC): flexible for complex and multi-factor payoffs.
- **PDEs**: numerical solution of valuation equations.

EQ: Valuation Models

Black-Scholes: Under the risk-neutral measure:

$$\frac{dS_t}{S_t} = (r - q)dt + \sigma dW_t$$

With q as the continuous dividend yield, the European call is:

$$C = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2)$$

$$d_1 = \frac{\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Put-call parity:

$$C - P = S_0 e^{-qT} - K e^{-rT}$$

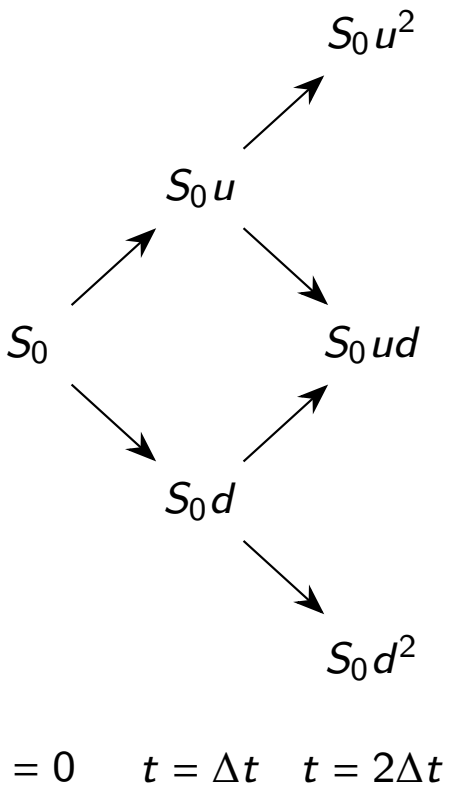
Binomial Tree: Basic Idea

Tree: At each short time step, the price can only:

- go up (multiply by u),
- or go down (multiply by d).

Step 1: build the price tree.
Step 2: compute the payoff at the end. For a call: $\max(S_T - K, 0)$.
Step 3: work backward by discounting:

$$V = e^{-r\Delta t}\left(p^* V_{\text{up}} + (1-p^*)V_{\text{down}}\right), \quad p^* = \frac{e^{(r-q)\Delta t} - d}{u-d}.$$



EQ: Valuation Models

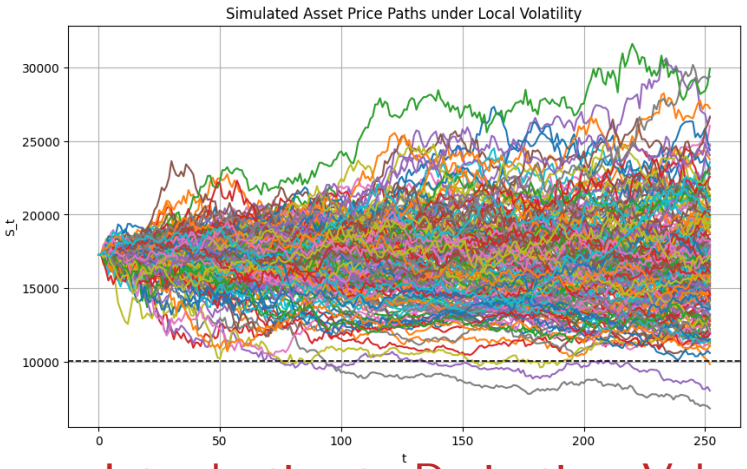
Monte Carlo

Price of a European derivative:

$$V_0 = e^{-rT} E^Q[\Phi(S_T)]$$

Paths are simulated, for example, under Black-Scholes:

$$S_T = S_0 \exp\left[\left(r - q - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\, Z\right], \qquad Z \sim N(0,1)$$



EQ: Valuation Models

PDEs (Black-Scholes)
For $V = V(S, t)$:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - q) S \frac{\partial V}{\partial S} - rV = 0$$

Terminal condition (European call):

$$V(S, T) = \max(S - K, 0)$$

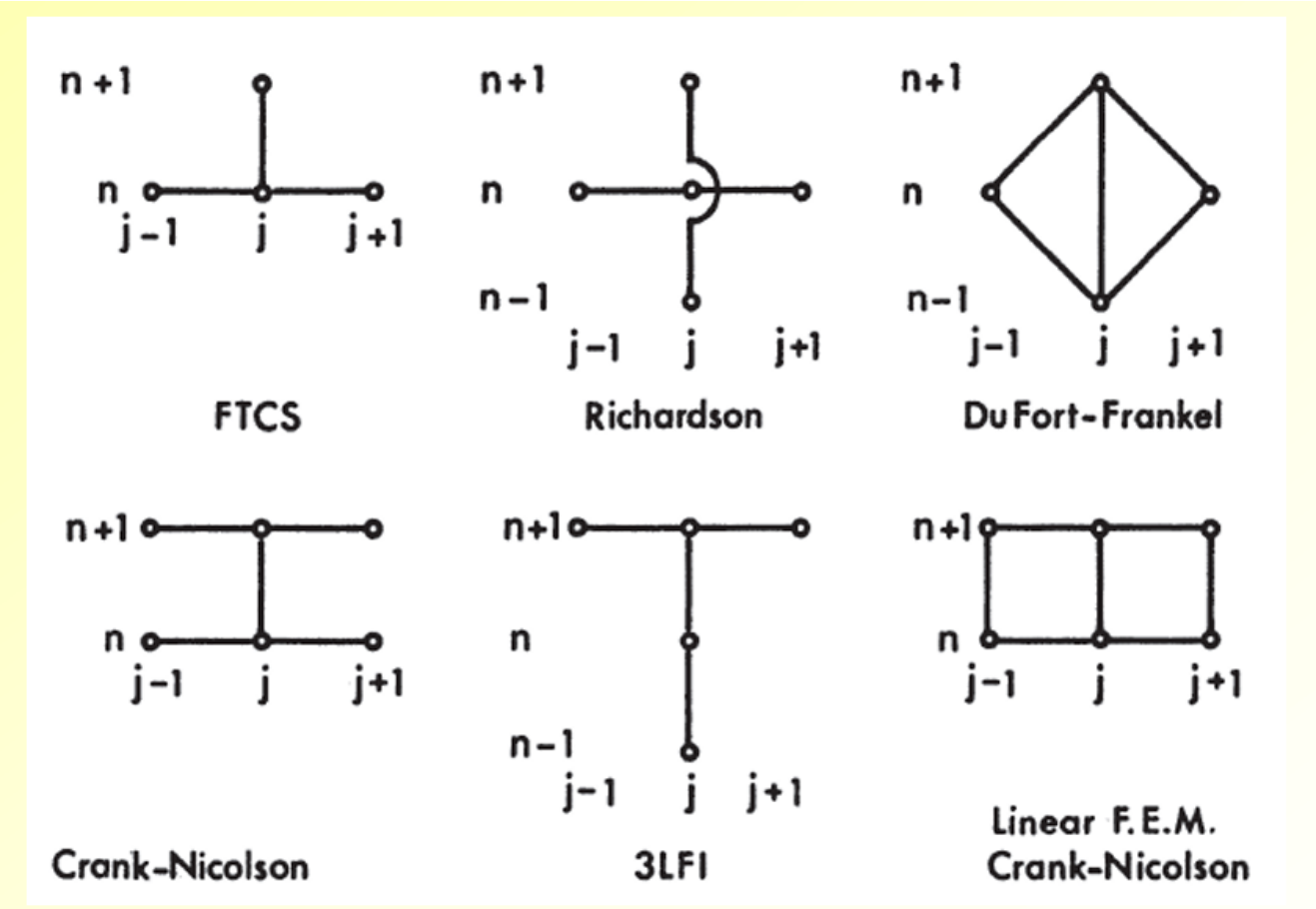
Typical boundaries:

$$V(0, t) = 0, \qquad V(S, t) \sim Se^{-q(T-t)} - Ke^{-r(T-t)} \quad \text{as } S \to \infty$$

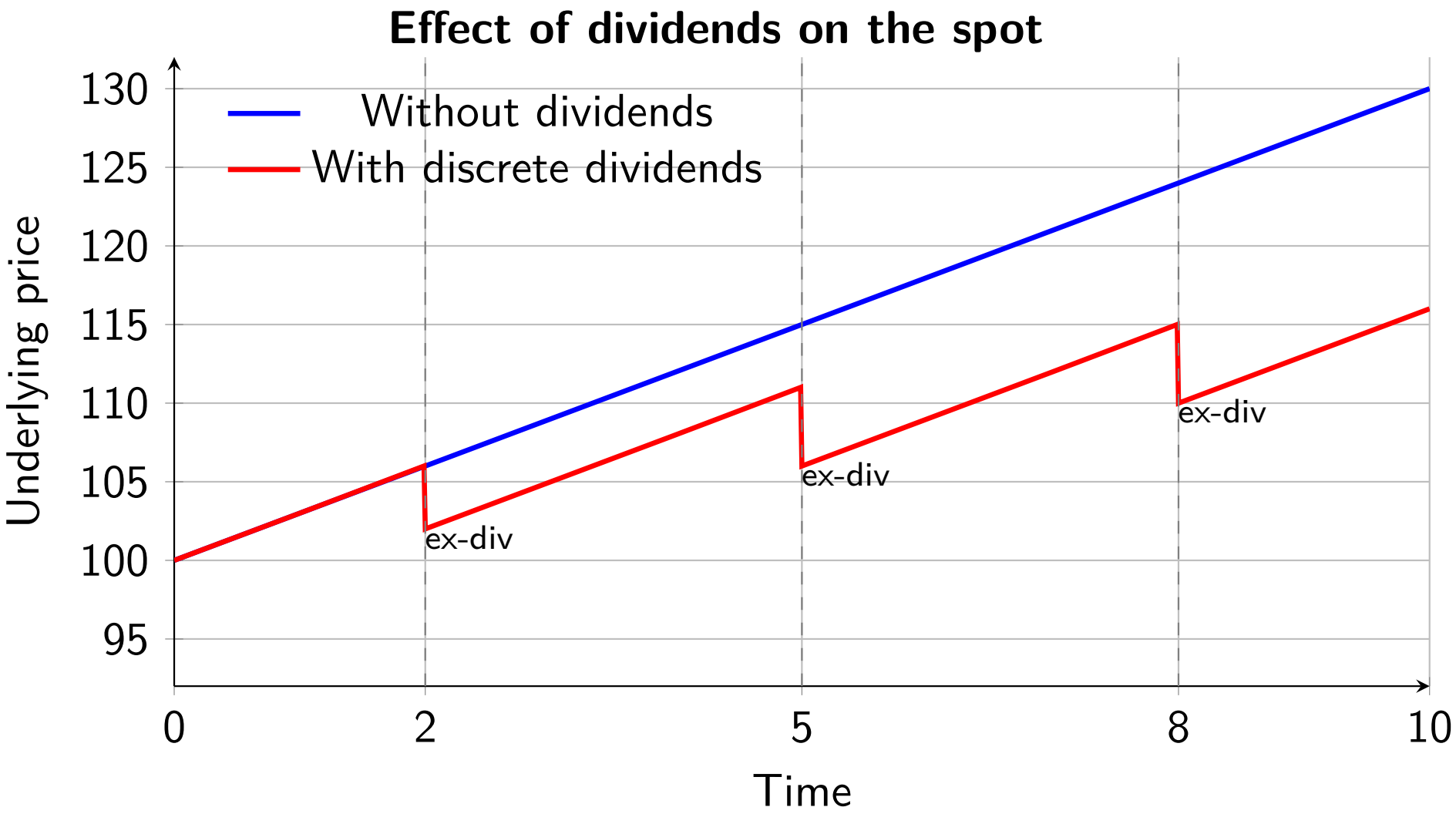
Methods: Crank-Nicolson, ADI, DuFort-Frankel, upwind, FTCS, ...

EQ: Valuation Models

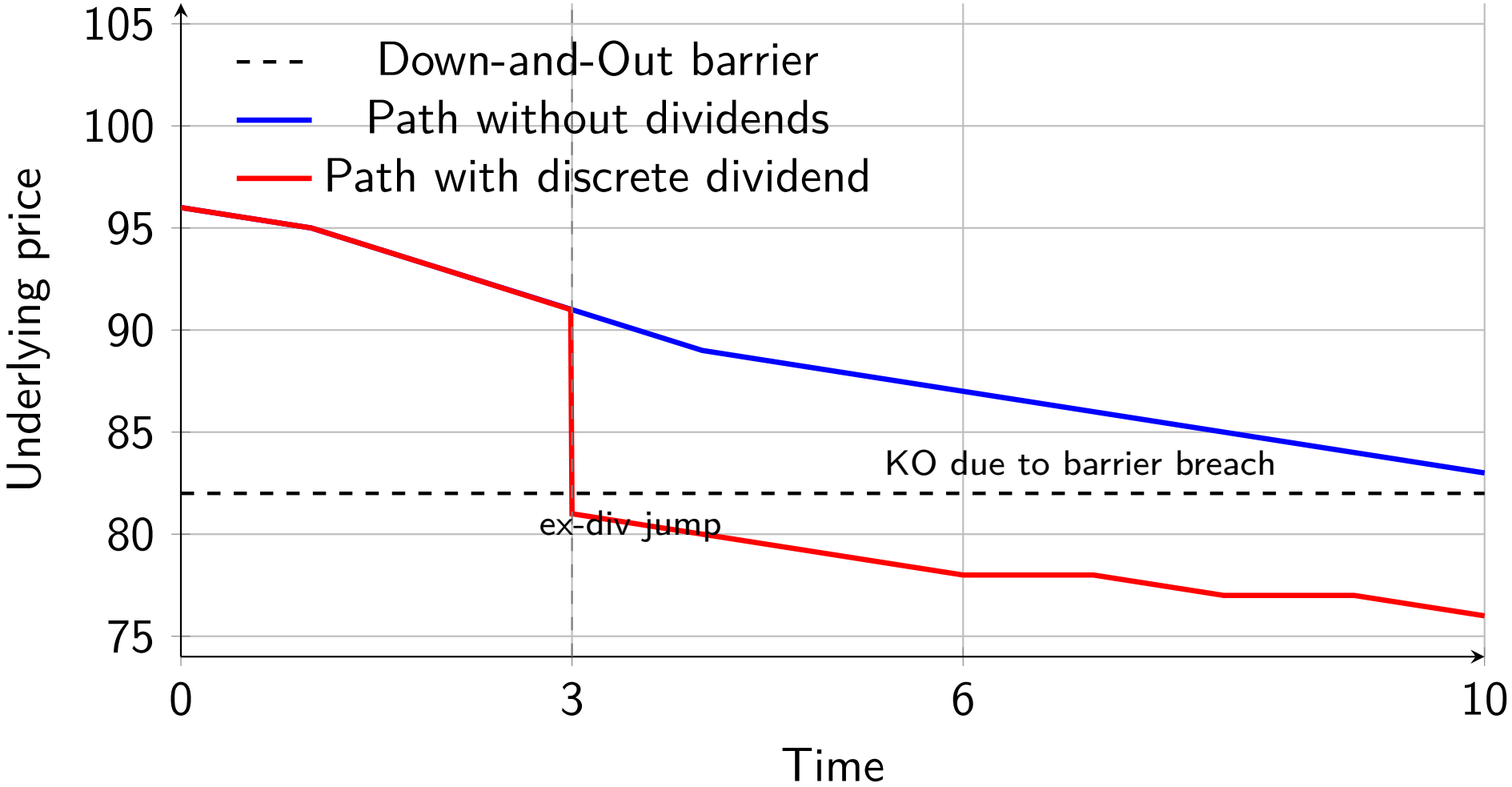
PDEs



EQ: Valuation Models



Impact of Discrete Dividends on a Barrier Option



EQ: Valuation Models

Limitations observed in the market:

- Volatility smile/skew.
- Heavier tails and jump events.
- Parameter instability over time.

Key message

- BS is the conceptual and operational foundation for vanilla options.
- For the real market, it is extended with smile models (Local Vol, Heston) and numerical methods (Tree, MC, PDE).
- Model choice must balance accuracy, robustness, and speed.

EQ: Smile, Surface and No-Arbitrage

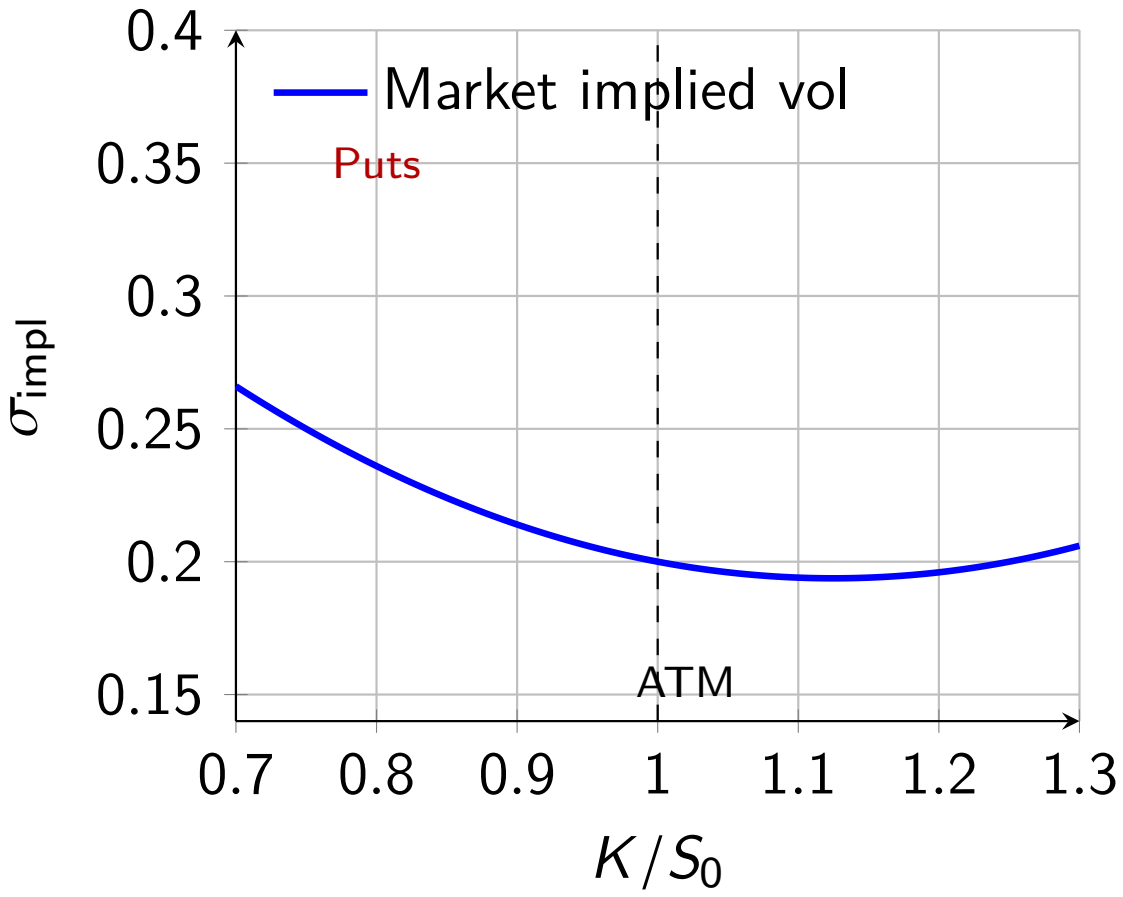
Implied volatility: from price to smile

- The market quotes option **prices**; **implied volatility** is the σ that reproduces that price in Black-Scholes.
- For a fixed maturity T, the function $K \mapsto \sigma_{\text{impl}}(K, T)$ defines the **smile/skew**.
- In equity, a **negative skew** is often observed: OTM puts are more expensive (in implied vol) than OTM calls.

$$\text{Market price} \xrightarrow{\text{BS inversion}} \sigma_{\text{impl}}(K, T)$$

EQ: Smile, Surface and No-Arbitrage

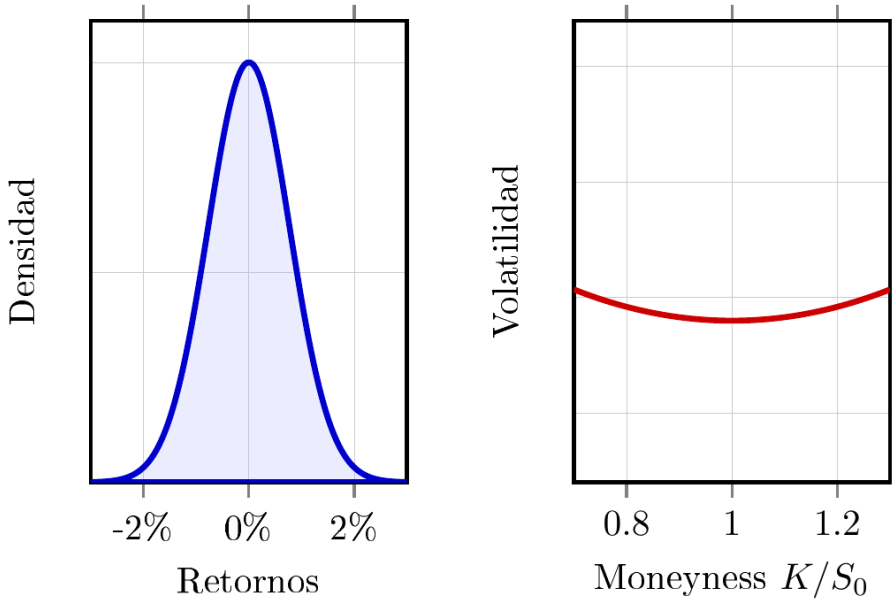
Smile/Skew in Equity: more expensive OTM puts



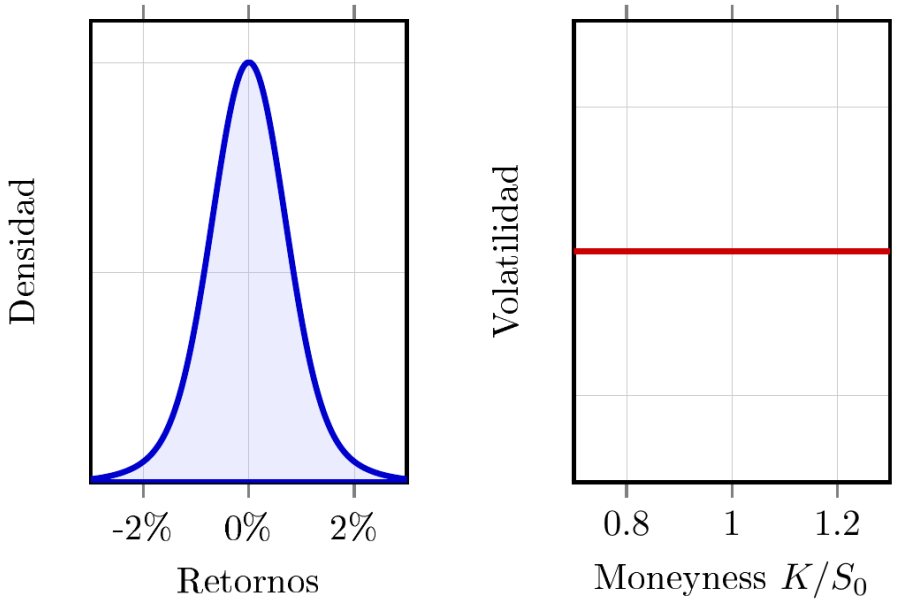
- **Left-tail risk**: sharp drops in the index/stock.
- **Structural demand for hedging**: investors buy puts for protection.
- **Leverage effect**: when spot falls, volatility rises.
- **Return asymmetry**: risk-neutral distribution skewed to the downside.

$$K/S_0 < 1 \;\Rightarrow\; \sigma_{\text{impl}} \uparrow$$

EQ: Smile, Surface and No-Arbitrage

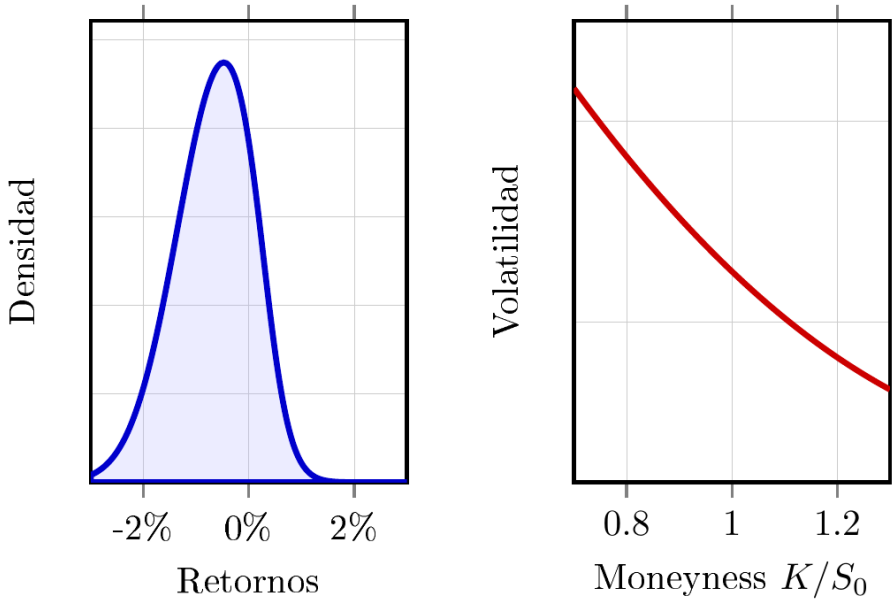


Scenario A

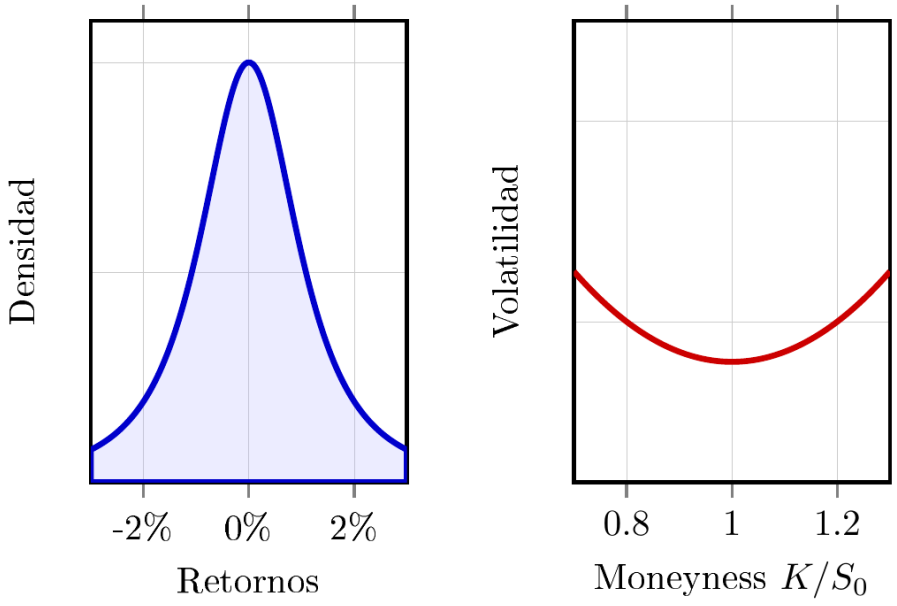


Flat vol

EQ: Smile, Surface and No-Arbitrage



Scenario B



Vol

EQ: Smile, Surface and No-Arbitrage

Volatility surface

- The implied surface is:

$$(K, T) \mapsto \sigma_{\text{impl}}(K, T)$$

- It is not enough to interpolate smoothly: it must be **consistent with no-arbitrage**.
- It is used to:
  - price vanillas away from quoted nodes,
  - calibrate models (Local Vol, Heston, LSV),
  - generate consistent Greeks.

EQ: Smile, Surface and No-Arbitrage

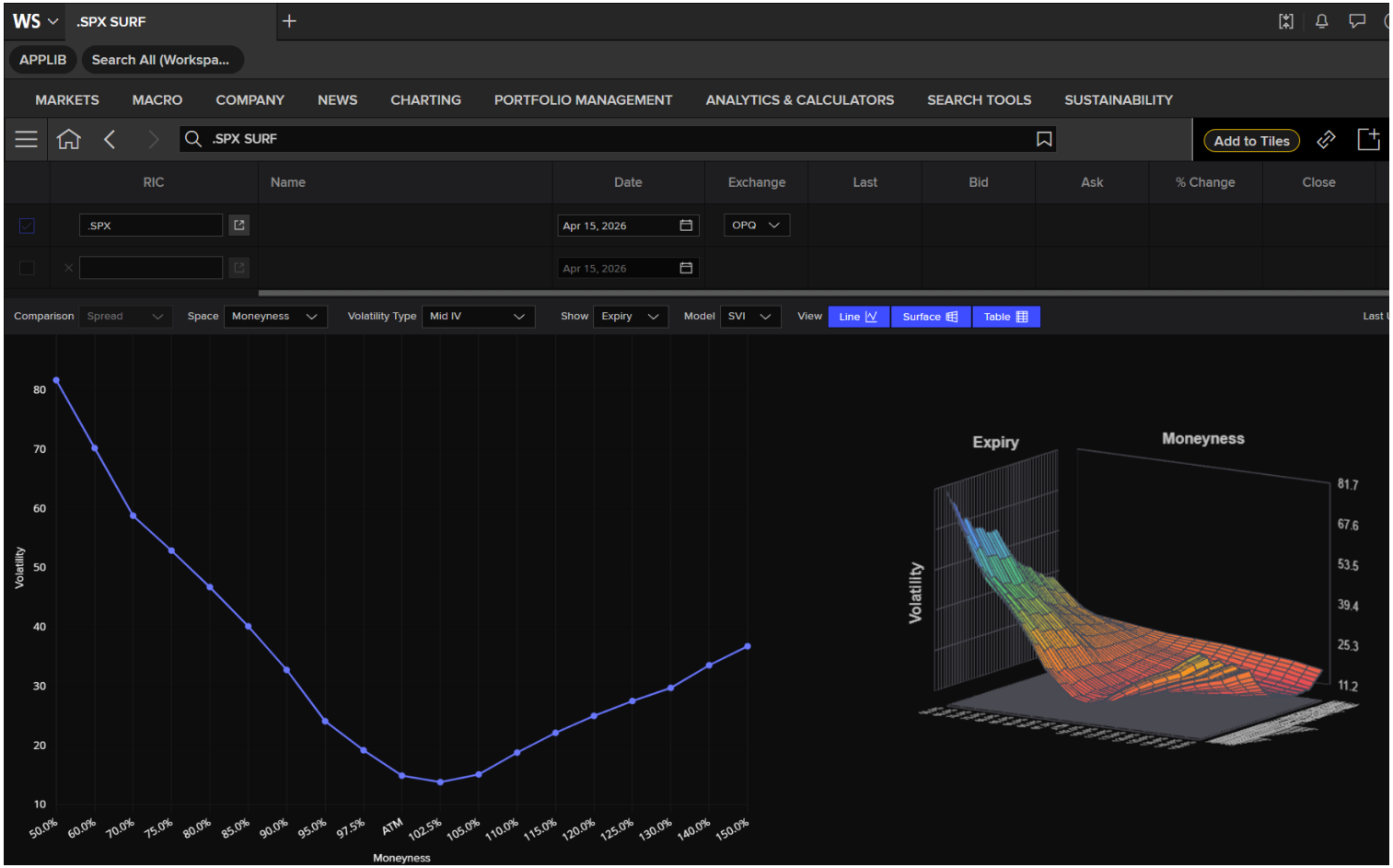


Figure: Volatility surface – Source: Reuters

EQ: Smile, Surface and No-Arbitrage

No-arbitrage on the surface We work with the European call price surface:

$$(K, T) \mapsto C(K, T)$$

The implied volatility surface must be consistent with this price surface.

Basic differential no-arbitrage conditions:

$$\frac{\partial C}{\partial K} \leq 0 \qquad \frac{\partial^2 C}{\partial K^2} \geq 0 \qquad \frac{\partial C}{\partial T} \geq 0$$

EQ: Smile, Surface and No-Arbitrage

Interpretation of ∂_K, ∂_{KK}, ∂_T.

1. Monotonicity in strike

$$\frac{\partial C}{\partial K} \leq 0$$

If the strike increases, the call price must decrease.

2. Convexity in strike (butterfly no-arbitrage)

$$\frac{\partial^2 C}{\partial K^2} \geq 0$$

It prevents butterfly arbitrage. In addition:

$$\frac{\partial^2 C}{\partial K^2} = f_{S_T}^{\mathbb{Q}}(K) \geq 0$$

3. Time monotonicity (calendar no-arbitrage)

$$\frac{\partial C}{\partial T} \geq 0$$

EQ: Advanced Models in Equity

Most common models in market practice:

- Black-Scholes: benchmark and common market language.
- Local Vol (Dupire): fits today's vanilla surface exactly.
- Heston (Stochastic Vol): stochastic volatility with more realistic skew/smile.
- Jump-Diffusion (Merton, Bates): incorporates jumps for tail behavior.
- Local-Stochastic Vol (LSV): combines vanilla fit with more realistic dynamics.

EQ: Advanced Models in Equity

Quick comparison

Model	Vanilla fit today	Smile dynamics	Computational cost
BS	Low (no smile)	None	Very low
Local Vol	Very high	Medium-low	Medium
Heston	High	High	Medium-high
LSV	Very high	High	High
Jump-Diffusion	Medium-high	High in the tails	Medium-high

Heston (I): Dynamics and Parameters

Under the risk-neutral measure:

$$dS_t = (r - q)S_t\,dt + \sqrt{v_t}\,S_t\,dW_t^S$$

$$dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v, \qquad dW_t^S\,dW_t^v = \rho\,dt$$

Parameter interpretation:

- θ: long-run variance.
- κ: mean-reversion speed.
- ξ: vol-of-vol (curvature/smile).
- ρ: spot-vol correlation (skew).
- v_0: initial variance.

Heston (II): Calibration, Advantages and Limits

Typical calibration:

- Objective: minimize the error between $\sigma_{impl}^{mkt}(K, T)$ and $\sigma_{impl}^{model}(K, T)$.
- Instruments: liquid vanillas across strikes and expiries.
- Useful constraint: Feller condition $2\kappa\theta \geq \xi^2$ (avoids hitting 0 too frequently).

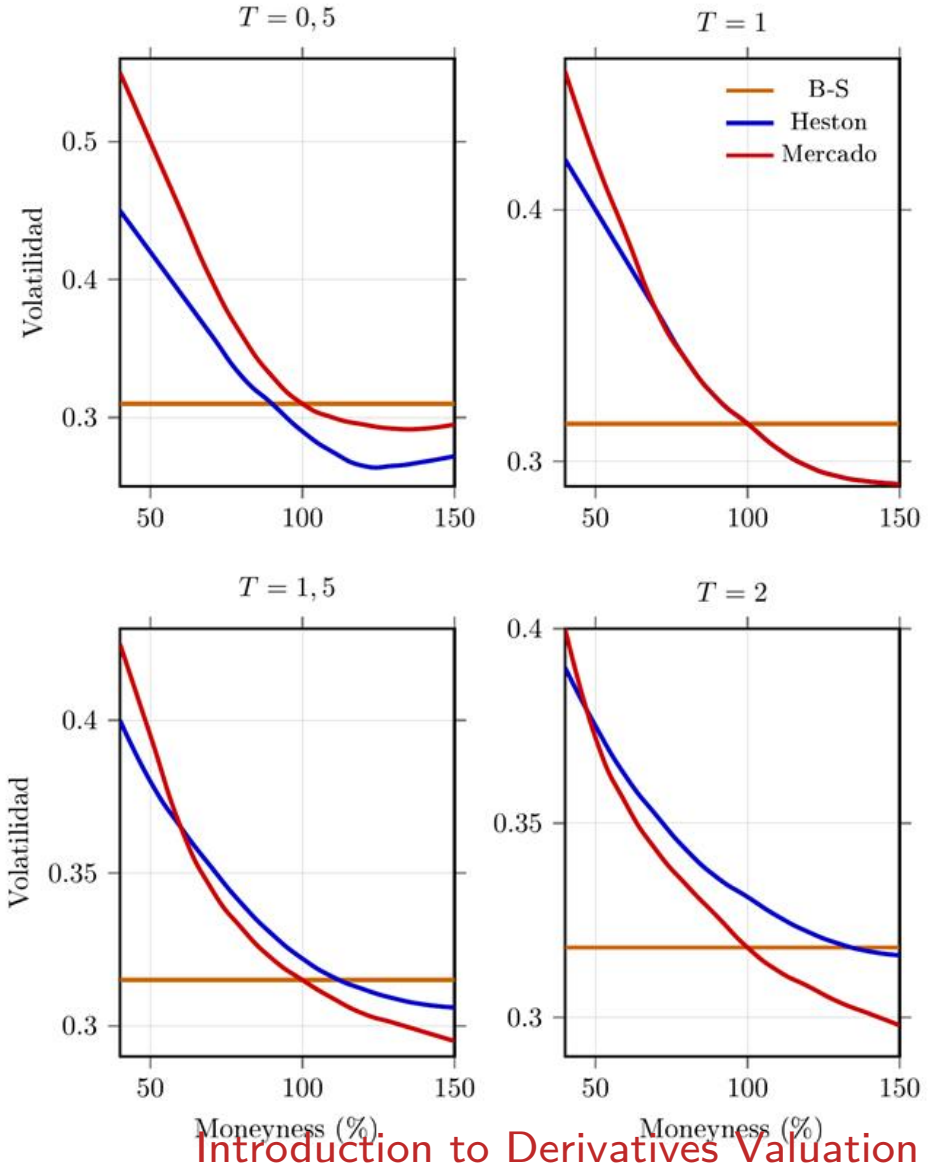
Advantages:

- Generates skew/smile structurally.
- Better hedge dynamics than pure Local Vol.
- Semi-closed form for vanillas via the characteristic function (fast to calibrate).

Limitations:

- It may not fit the entire surface perfectly. It is calibrated to a specific T.
- Sensitive to parameter stability across days.

Heston (III): Intuition



Local Vol (I): Idea and Dupire Equation

Volatility becomes a deterministic function of (S, t).

$$dS_t = (r - q)S_t\, dt + \sigma_{loc}(S_t, t)\, S_t\, dW_t$$

If $C(K, T)$ is smooth and arbitrage-free. **Dupire equation**:

$$\sigma^2_{loc}(K, T) = \frac{\frac{\partial C}{\partial T} + (r - q)K\frac{\partial C}{\partial K} + qC}{\frac{1}{2}K^2\frac{\partial^2 C}{\partial K^2}}$$

Local Vol (II)

Practical pipeline:

- The input is the EQ implied volatility surface.
- Interpolate/smooth the surface $C(K, T)$ in an arbitrage-free way (Hagan interpolator).
- Compute derivatives $\partial_T C$, $\partial_K C$, $\partial_{KK} C$.
- Obtain $\sigma_{loc}(K, T)$ and solve PDE/MC for pricing.

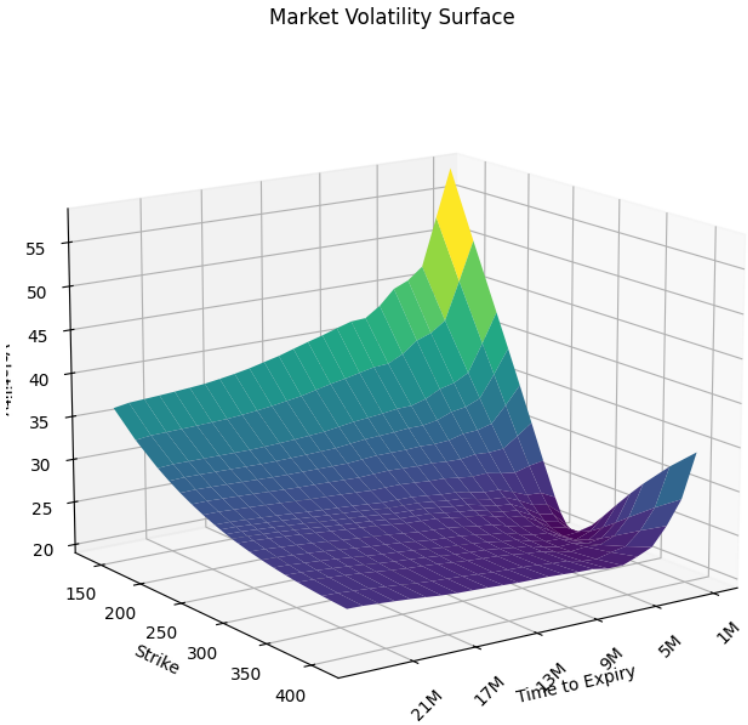
Advantages:

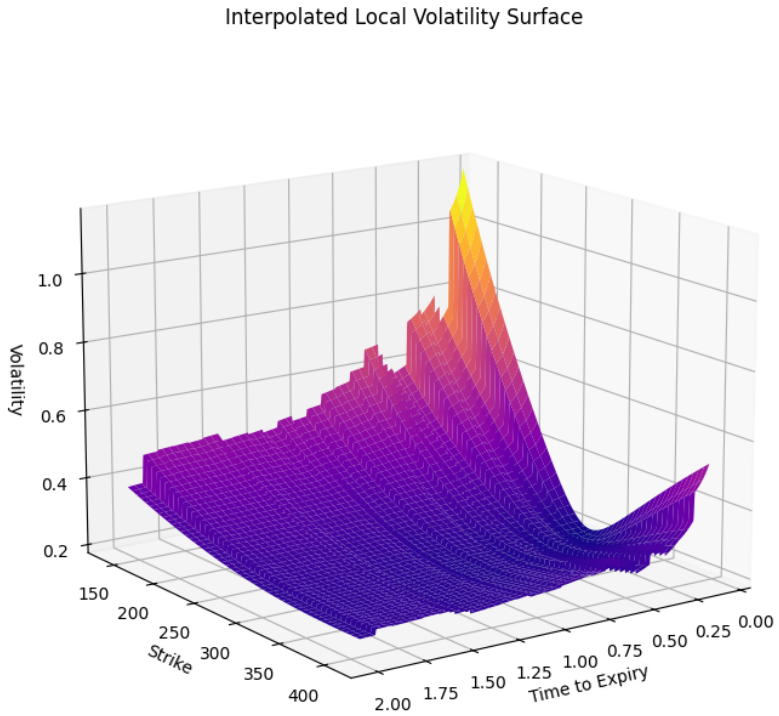
- Exact vanilla fit by construction.
- Good framework for static no-arbitrage control.

Limitations:

- Numerical derivatives are unstable if the grid is not properly regularized (a fairly difficult problem).
- Future smile dynamics are less realistic for certain exotics.

Local Vol (III): Intuition





The figure compares an implied volatility surface with the local volatility surface extracted from it. The local volatility surface is the deterministic volatility function implied by the option price surface through Dupire's formula.

IR: Introduction and Differences vs EQ

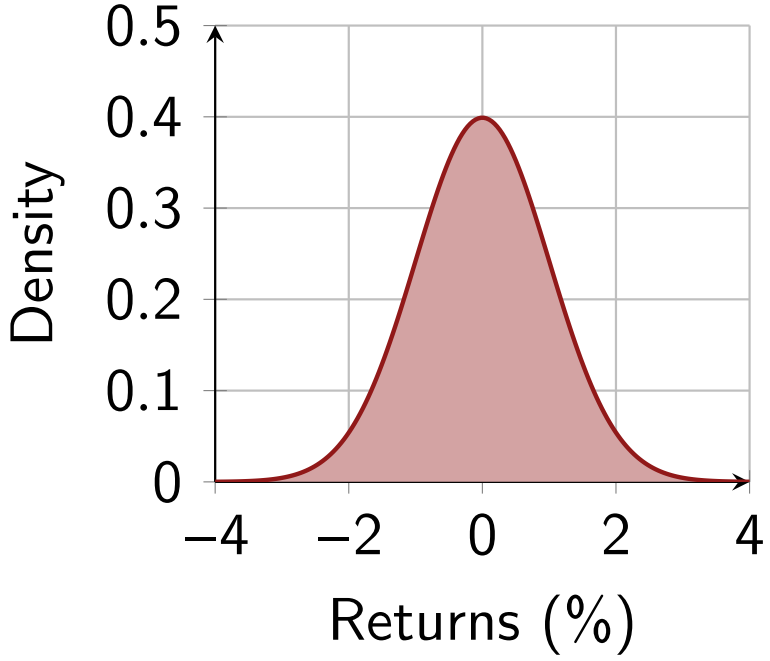
IR complexities:

- **Dimensionality**: in EQ the underlying is a point (one-dimensional), in IR the underlying is a curve (multi-dimensional).
- **Rates and Curves**: in IR there are many interest rates (spot, forward, swap, etc.) and curves (spot, forward, discount ...).
- **Multi-curve**: in a multi-curve environment (ESTR, EUR1M, EUR3M, ...), each curve has its own dynamics and volatility.
- **Notation**: in EQ we speak about S_t, while in IR we speak about $R(t, T)$ where t is the current time and T is the tenor.
- **More complex models**: IR curve dynamics usually require more sophisticated models (LMM, HW) than EQ.
- **More varied instruments**: in IR there is a wide variety of vanilla and exotic derivatives, more complex than in EQ.

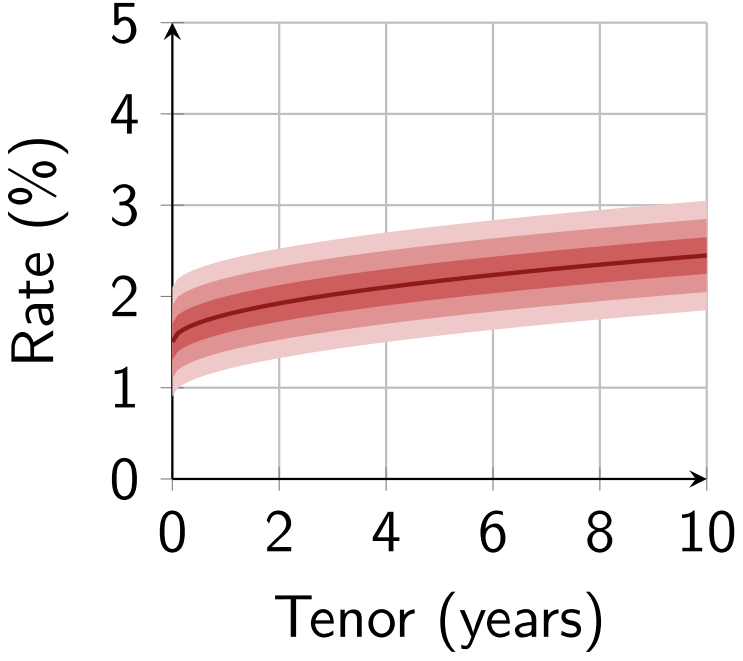
IR: Introduction and Differences vs EQ

EQ vs IR distributions for the same expiry:

EQ: One distribution



IR: Infinite distributions



While in EQ there is only one distribution for each expiry, in IR there is one distribution for each point T on the curve, that is, infinitely many distributions.

IR: Introduction and Differences vs EQ

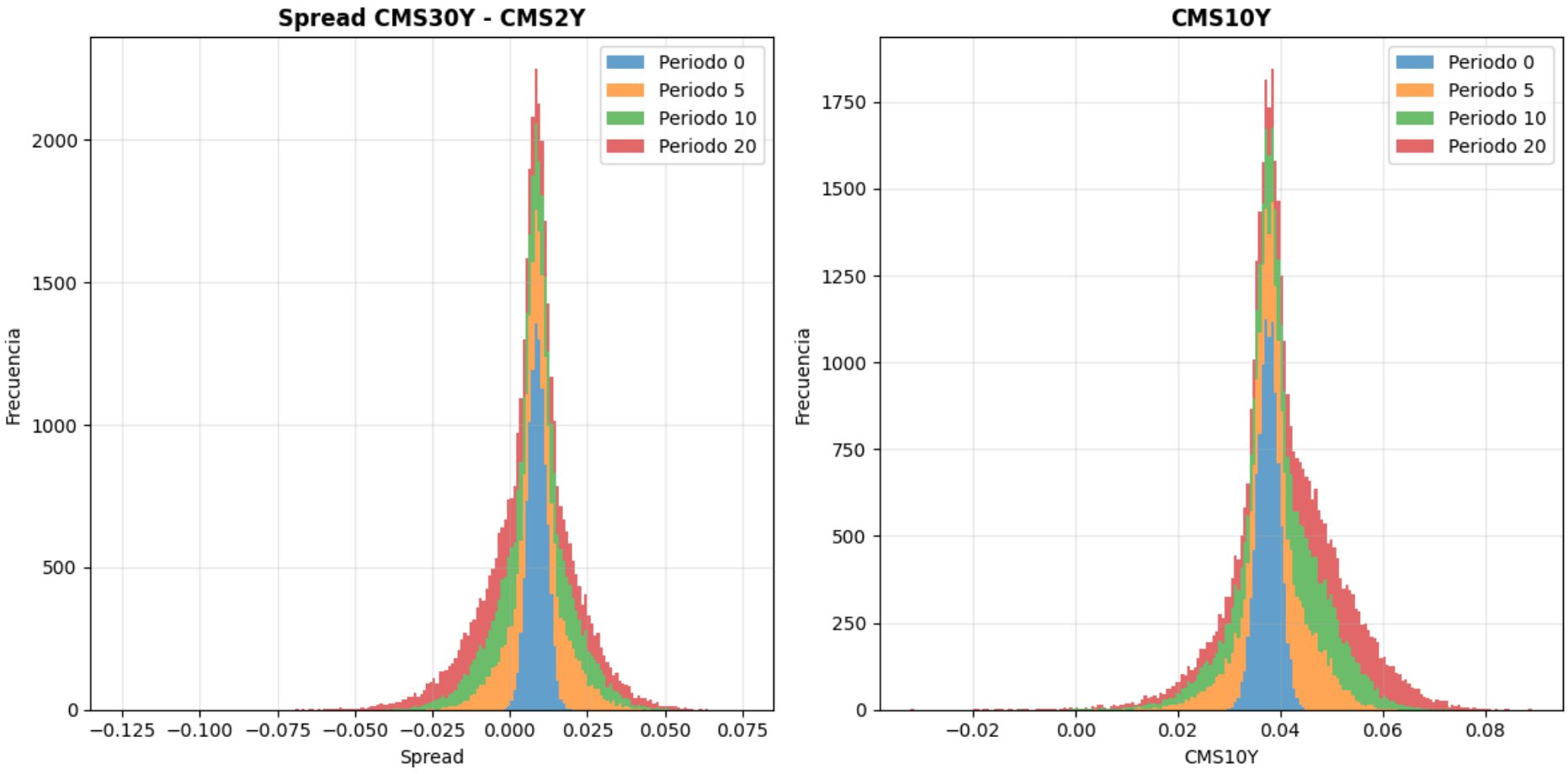


Figure: Each point T on the curve has its own distribution, for several points in time, t.

Rates and Curves

In IR we have a wide variety of rates and curves. All of them refer to the same concept: the time value of money:

Interest Rates:

- **Spot**: interest rate for a period starting at the current time, $R(t, T)$.
- **Forward**: interest rate for a period starting in the future, $F(t; T, S)$.
- **Instantaneous Spot**: interest rate for an infinitesimal period starting at the current time, $r(t)$.
- **Instantaneous Forward**: interest rate for an infinitesimal period starting in the future, $f(t, T)$.
- **Zero-Coupon Bond**: present value of 1 unit of money received in the future, $P(t, T)$.

Rates and Curves

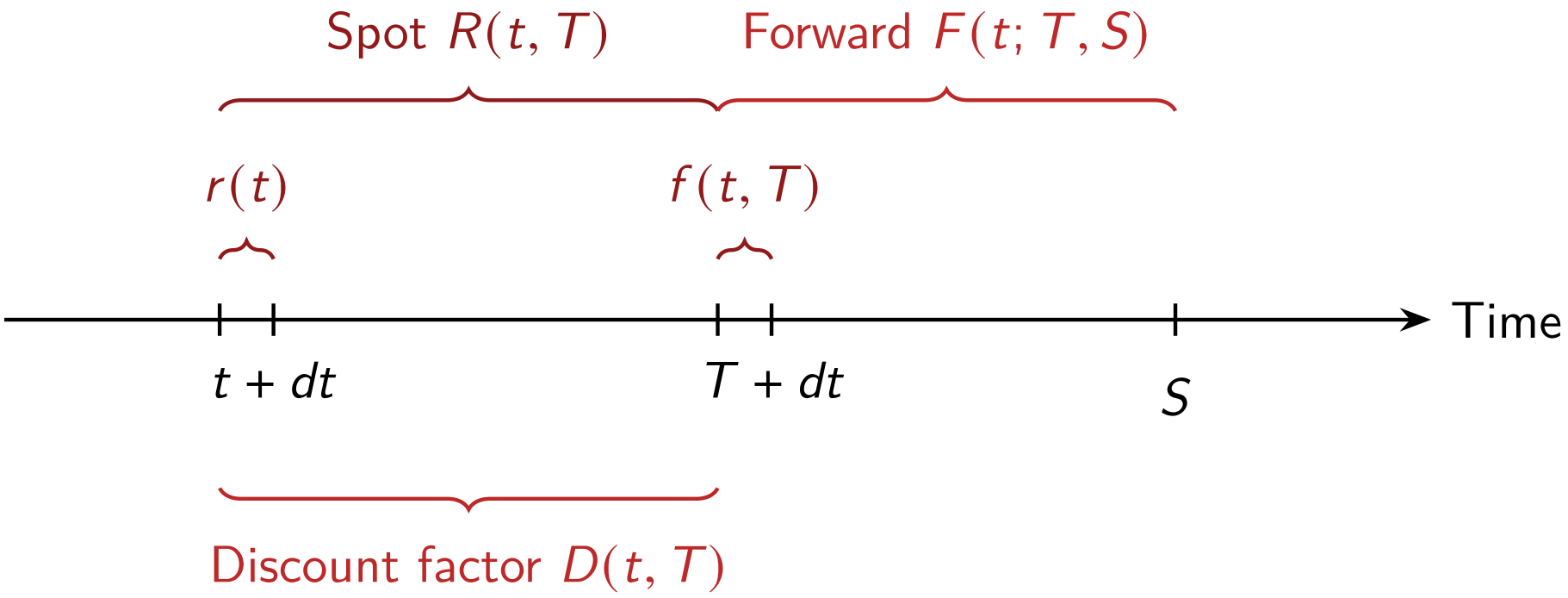
- **Discount Factor**: present value of 1 unit of money received in the future, $D(t, T)$.
- **Swap**: weighted average of forward rates for a given tenor, starting at T_α and ending at T_β, $S(t; T_\alpha, T_\beta)$.
- **CMS**: swap rate for a given constant tenor (for example 10 years or 30 years).

When rates are deterministic (non-stochastic), we have:

$$P(t, T) = D(t, T)$$

Note: at $t = 0$, all variables are market-observable, but for $t > 0$ they are random and not observable.

Rates and Curves



The current time is denoted by t and can be the present $t = 0$ or the future $t > 0$.

Rates and Curves

Interest Rate Curves:

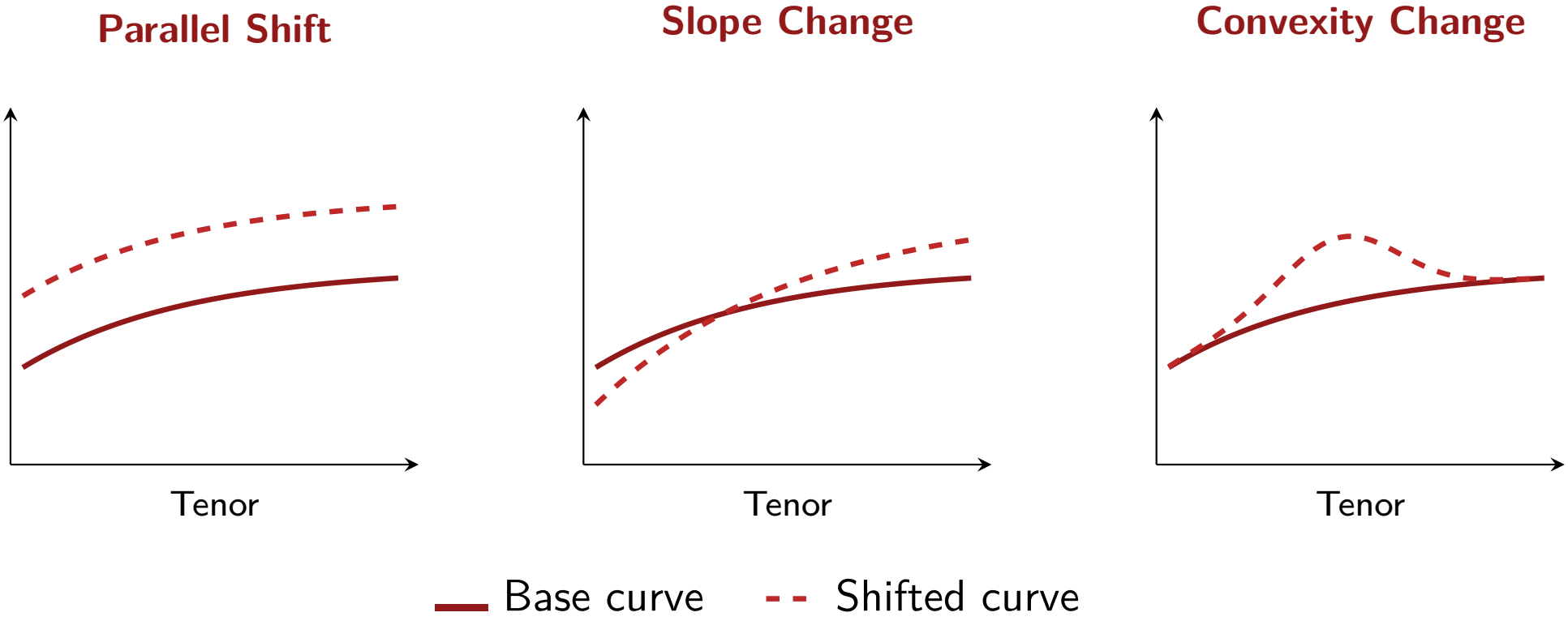
- **Spot**: function $T \mapsto R(t, T)$.
- **Forward**: given $\Delta t = S - T$, function $T \mapsto F(t; T, T + \Delta t)$.
- **Discount**: function $T \mapsto P(t, T)$.

Curves at $t = 0$ are market-observable and are usually denoted with superscript $\mathcal{M}$:

$$R^{\mathcal{M}}(0, T), \quad F^{\mathcal{M}}(0; T, S), \quad P^{\mathcal{M}}(0, T)$$

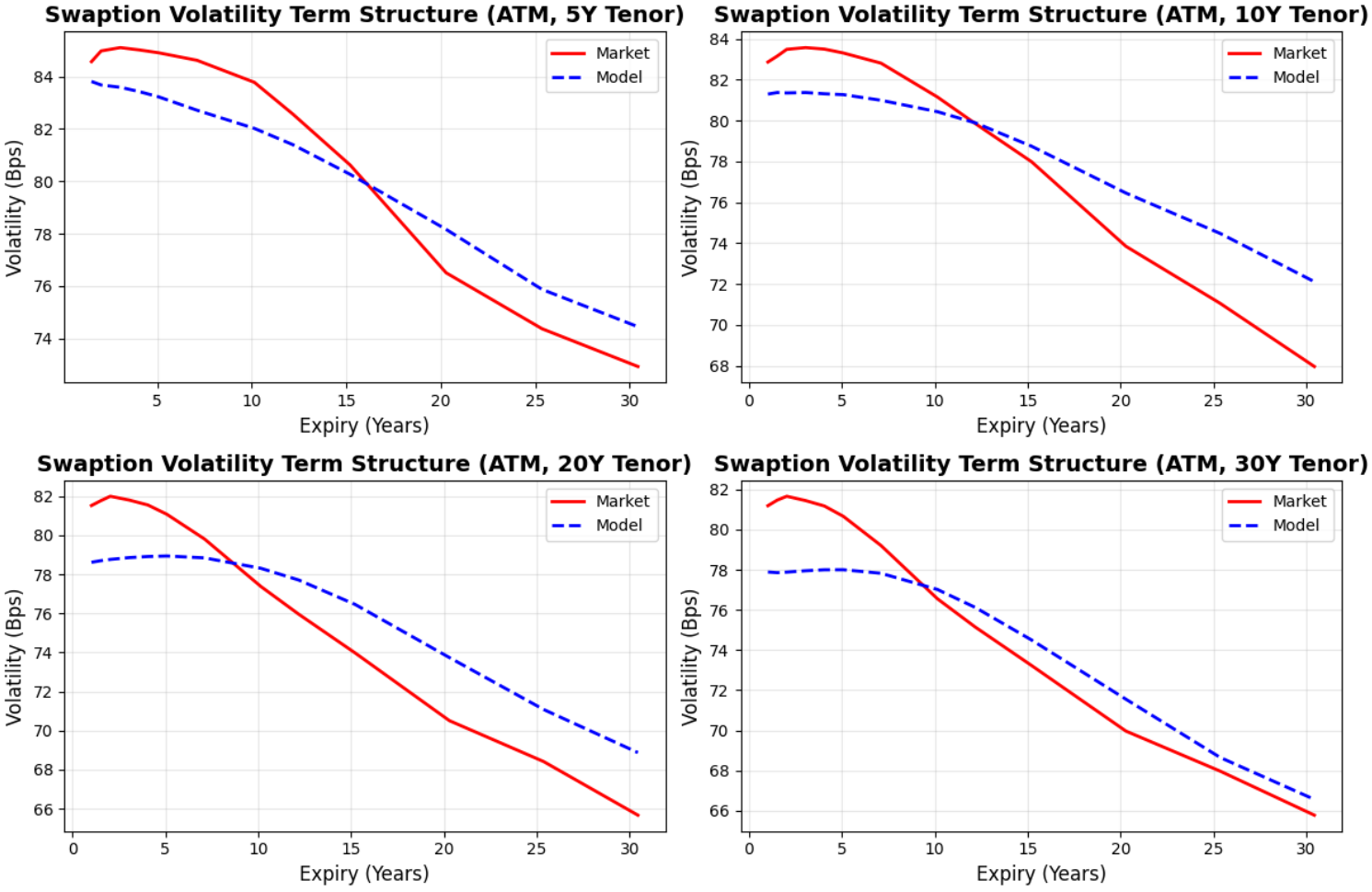
Note: for $t > 0$, the curves are random and not observable and need to be constructed with a model.

Rates and Curves: Curve Movements



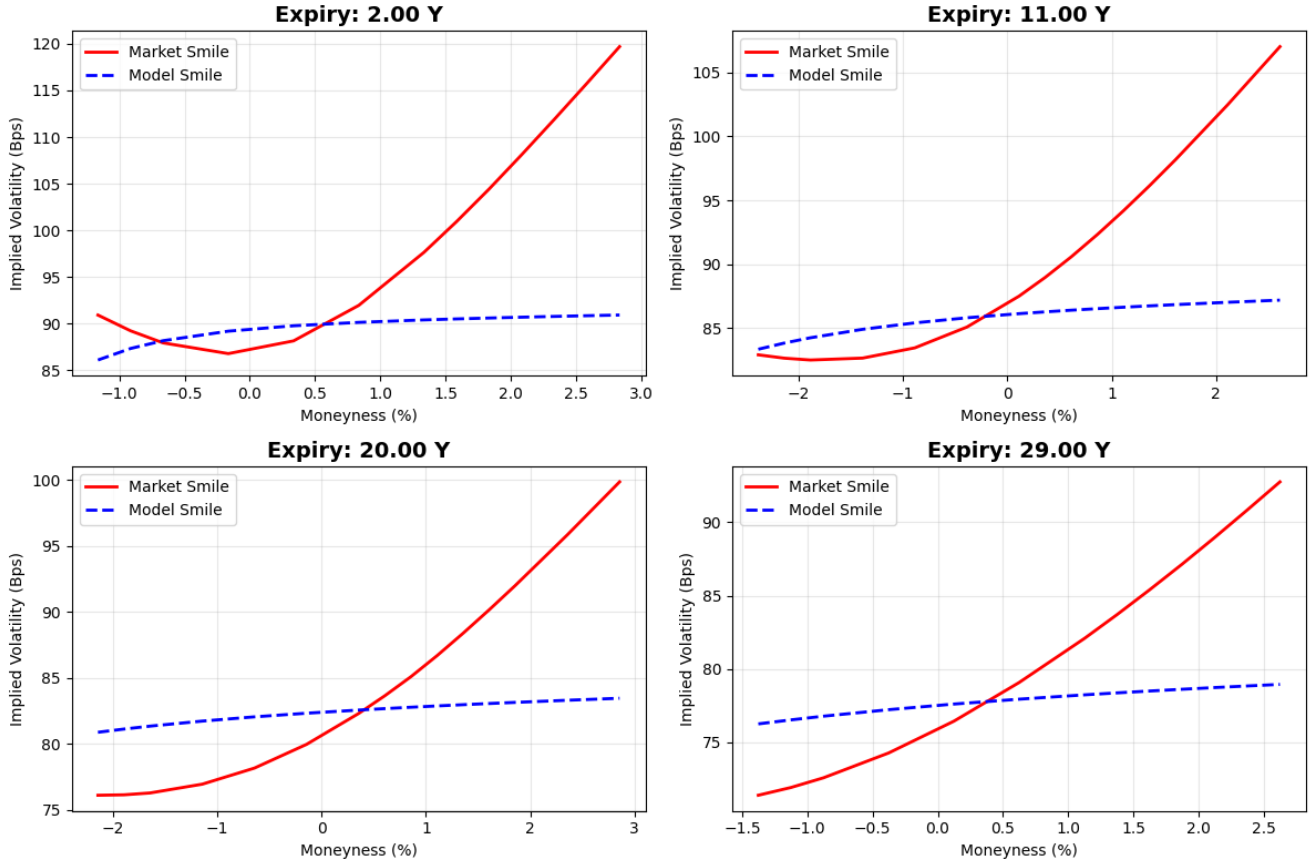
The vast majority of curve movements can be explained by changes in level and slope.

Volatility Term Structure



The goal of IR models is to capture the term structure of volatility, which usually presents a humped shape.

Volatility Smile in IR



We also have volatility smile in IR. However, it is very challenging to capture both the term structure of volatility and the smile at the same time, and a different model is needed for each.

IR Instruments

Vanilla:

- **Bonds**: zero-coupon or coupon-bearing bonds.
- **FRNs**: floating-rate notes referenced to an interest rate (for example, EUR3M).
- **FRAs**: contracts to buy or sell a future interest rate on a future date.
- **Swaps**: exchange of interest cash flows between two parties, for example a fixed-for-floating swap.
- **Caps/Floors**: options on future interest rates (forward rates).
- **Swaptions**: options on swaps (options on swap rates).

IR Instruments

Exotics:

- **CMS**: swap that pays a swap rate (for example, 10 years) on the floating leg instead of a forward rate.
- **CMS Spread**: instrument that pays the difference between two swap rates (for example, 10 years minus 2 years).
- **CMS Cap/Floor**: option on a CMS rate. There is also a CMS Spread Cap/Floor version.
- **Range Accruals**: bond that accrues interest only when the interest rate stays within a given range.
- **Bermudan Swaption**: options on swaps that can be exercised on multiple dates (for example, every 6 months).
- **Structured Notes**: bonds that pay a coupon linked to a complex formula that may include interest rates, indices, etc.

Black76 vs Bachelier

Feature	Black76	Bachelier
FWD dynamics	$dF = \sigma F dW_t$	$dF = \sigma dW_t$
SWPN dynamics	$dS = \sigma S dW_t$	$dS = \sigma dW_t$
Noise type	Log-normal	Normal
Soporte	$F, S > 0$	$F, S \in \mathbb{R}$
Typical use	Caps/Floors, Swpn	Caps/Floors, Swpn
Smile/Skew	No	No
Term structure	No	No
Speed	Very high	Very high
Volatility	Log-normal implied	Normal implied

Where $F = F(t; T, S)$ and $S = S(t; T_\alpha, T_\beta)$.

Caplets: Black76 and Bachelier

Black76 (caplet, log-normal):

$$\text{Caplet}^{Black} = e^{-rT}\,\Delta t\,\left[F\,\Phi(d_1) - K\,\Phi(d_2)\right]$$

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Bachelier (caplet, normal):

$$\text{Caplet}^{Bach} = e^{-rT}\,\Delta t\,\left[(F-K)\,\Phi(d) + \sigma\sqrt{T}\,\varphi(d)\right]$$

$$d = \frac{F-K}{\sigma\sqrt{T}}$$

Where $\Phi(\cdot)$ is the CDF and $\varphi(\cdot)$ the PDF of the standard normal distribution. F is the forward, K the strike, T the time to maturity, r the risk-free rate, and Δt the year fraction.

Swaptions: Black76 and Bachelier

Black76 (log-normal):

$$\text{Swaption}^{Black} = e^{-rT}\, A\, [S\, \Phi(d_1) - K\, \Phi(d_2)]$$

$$d_1 = \frac{\ln(S/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Bachelier (normal):

$$\text{Swaption}^{Bach} = e^{-rT}\, A \left[(S - K)\, \Phi(d) + \sigma\sqrt{T}\, \varphi(d) \right]$$

$$d = \frac{S - K}{\sigma\sqrt{T}}$$

Where $\Phi(\cdot)$ is the CDF and $\varphi(\cdot)$ the PDF of the standard normal distribution. S is the swap rate, K the strike, T the time to maturity, r the risk-free rate, and A the swap annuity.

IR: Models

Quick comparison:

Model	Term Structure	Smile	Computational cost
Black76	No	No	Very low
Bachelier	No	No	Very low
HW 1F	Yes	No	Medium
HW 2F	Yes	Partial	High
LMM	Yes	No	Low
SABR	No	Yes	Medium

Term Structure in IR: Hull-White 1F

Black76 and Bachelier models have several limitations that make them poorly suited to modeling the curve as a whole:

- They do not model the full curve, only a single point (a forward rate or a swap rate).
- They assume normal or lognormal dynamics, which is not realistic for interest rates (mean reversion).
- They cannot capture the volatility term structure, which is crucial for pricing derivatives with different expiries.
- There is no correlation between different points on the curve.

Black76 and Bachelier are a pragmatic workaround that allows us to price certain simple IRDs (caps/floors and swaptions) in a Black-Scholes-like way, but they are insufficient for modeling the curve or exotic derivatives.

Term Structure in IR: Hull-White 1F

The simplest model with a full term structure is the **one-factor Hull-White model (HW 1F)**. It works as follows:

1. We simulate the shortest point on the curve, the instantaneous spot rate $r(t)$.
2. From $r(t)$, stochastic discount factors $D(t, T)$ are computed for all times t and tenors T.
3. With $D(t, T)$ we obtain zero-coupon bond prices $P(t, T)$.
4. Zero-coupon bonds allow us to construct any rate curve (discount, spot, forward).
5. With the full underlying (curve) and its dynamics, we can price any IRD, from vanillas to exotics.

Term Structure in IR: Hull-White 1F

1. Instantaneous spot rate dynamics:

Under the risk-neutral measure, the instantaneous spot rate $r(t)$ follows mean-reverting dynamics:

$$dr(t) = \kappa(\theta(t) - r(t))dt + \sigma dW(t)$$

Where:

- κ: mean-reversion speed.
- $\theta(t)$: long-run level, allowing the model to fit the initial rate curve.
- σ: interest-rate volatility.
- $W(t)$: standard Brownian motion.

Term Structure in IR: Hull-White 1F

Several important conclusions follow from the expression above:

- The interest rate $r(t)$ is stochastic and normally distributed (it can be negative).
- $\theta(t)$ is a deterministic function of time that must be fitted so the model reproduces the initial rate curve.
- The shock driven by $dW(t)$ is the same for all tenors, implying correlation 1 across all points on the curve.
- Volatility σ is constant, which prevents the model from capturing the volatility smile.

Term Structure in IR: Hull-White 1F

2. Stochastic discount factor:

From the dynamics of $r(t)$, we can derive the expression for the stochastic discount factor $D(t, T)$:

$$D(t, T) = e^{-\int_t^T r(s)ds}$$

Suppose the short rate were constant and equal to r. In that case:

$$D(t, T) = e^{-r(T-t)}$$

And if discounting starts at $t = 0$:

$$D(0, T) = e^{-rT}$$

This is the constant-rate discounting formula used in EQ.

Term Structure in IR: Hull-White 1F

3. Zero-coupon bonds:

A zero-coupon bond is a derivative whose payoff at maturity is:

$$P(T, T) = 1$$

Now recall the fundamental valuation formula under the risk-neutral measure:

$$V(t) = \mathbb{E}^{\mathbb{Q}}\left[D(t, T)V(T) \mid \mathcal{F}_t\right]$$

Applying this formula to the zero-coupon bond, we have:

$$P(t, T) = \mathbb{E}^{\mathbb{Q}}\left[D(t, T)P(T, T) \mid \mathcal{F}_t\right] = \mathbb{E}^{\mathbb{Q}}\left[D(t, T) \mid \mathcal{F}_t\right]$$

The price of a derivative is the expected value of its discounted payoff at the current time (which may be $t = 0$ or $t > 0$). In IR, discounting is stochastic and is included inside the expectation.

Term Structure in IR: Hull-White 1F

That is, the zero-coupon bond price $P(t, T)$ is the expected value of the stochastic discount factor $D(t, T)$ under the risk-neutral measure. At $t = 0$, we have:

$$P(0, T) = \mathbb{E}^{\mathbb{Q}}\left[D(0, T)\right] = P^{\mathcal{M}}(0, T)$$

which is the market zero-coupon bond curve.

Note: $P(0, T)$ is market-observable, hence deterministic. However, $P(t, T)$ for $t > 0$ is random and not observable, and is obtained through the model.

Term Structure in IR: Hull-White 1F

4. Curves:

From zero-coupon bonds $P(t, T)$, all present and future interest-rate curves can be constructed:

- Discount curve: $P(t, T)$.
- Spot curve: $R(t, T) = -\frac{1}{T-t} \ln P(t, T)$.
- Forward curve: $F(t; T, S) = \frac{1}{S-T}(\frac{P(t,T)}{P(t,S)} - 1)$.
- Instantaneous forward curve: $f(t, T) = -\frac{\partial}{\partial T} \ln P(t, T)$.
- Swap curve: $S(t; T_\alpha, T_\beta) = \frac{P(t,T_\alpha) - P(t,T_\beta)}{\sum_i \Delta t_i P(t,T_i)}$.

However, remember that all these curves have correlation 1 across their points. **We can only generate parallel-shift scenarios** (no slope or convexity changes).

Term Structure in IR: Hull-White 1F

5. IRD:

With the full underlying (curve) and its dynamics, we can price any IRD, from vanillas to exotics. For example, for a caplet:

$$\text{Caplet}^{HW} = \mathbb{E}^{\mathbb{Q}}\left[D(0,T)F(T;T,S)\right]$$

Or derivatives whose payoff function, g, depends on the future spot rate (at time T) for tenor S, for example:

$$V(0) = \mathbb{E}^{\mathbb{Q}}\left[D(0,T)g(R(T,S))\right]$$

The only derivatives we could not price would be early-exercise derivatives (Bermudans, Americans) or derivatives on spreads or curve convexity.

In some cases there are closed-form formulas for IRDs in the HW model, but if not, we can always compute the derivative value with Monte Carlo.